

Step 2: Frame — Analysis Brief

CAPSTONE-01 | AI-Assisted Audit Control Framework · MCX AI Lab: Data Analysis with AI
Dataset: grc_risk_register.csv — synthetic GRC risk register, 160 rows × 19 columns · snapshot date 30 Jun 2026

1 PURPOSE AND AUDIENCE

This brief fixes the analytical question before any analysis is run, so that findings in Step 3 answer a question defined in advance rather than one selected after the fact. The deliverable is an **analysis** of the remediation inventory — what is true, what is uncertain, what warrants a closer look. It is not a remediation plan: mitigation decisions, tolerance thresholds and guardrails sit with senior leadership once findings are presented.

Audience	Audit Committee and senior leadership, at a quarterly risk and remediation review. Single artifact, two densities: a ten-second top layer for the leadership view, an exception-and-evidence layer beneath it for the committee.
Rationale	The committee version is the harder constraint and converts cheaply into a regulator or CRO briefing; the leadership view is a subset of it, not a separate document.

2 FRAMING DECISIONS LOCKED

Q1 Decision frame	The aged-item inventory question — what the unresolved population looks like and how it behaves. Analysis only; forward-looking content is labelled separately as an implication for management and never pre-decides the response.
Q2 Canonical metric	Overdue-to-SLA ratio — days past due divided by the original SLA window (Due_Date minus Date_Identified), with absolute days past due alongside. It normalises for severity-differentiated SLA windows and never touches Risk_Tier, so it survives if Step 4 confirms tier is a generator artifact.
Q3 Population scope	All 160 items. Closed items are retained as the control group that gives the unresolved population meaning.
Q4 Comparison basis	Cross-sectional, not temporal. Segments against each other and against the eligible-population mean (department, risk domain, owner role, tier, identification source), plus the closed-versus-breached contrast. No trend claims. No external benchmark — none exists in the file and none will be invented.
Q5 Constraints	Standard audit-committee action space: accept, direct, escalate, commission further testing. Context for how far an observation may lean, not a menu the analysis selects from. Observations implying management-level actions (reassigning owners, resetting SLAs) are out of bounds here.
Q6 Success test	The brief works if a senior leader finishes it knowing what is true, what is uncertain, and where their decision needs to land — with no decision pre-made for them. Quality gate for every downstream finding: if a sentence makes the decision rather than informing it, it is cut.

3 METRIC DEFINITIONS AND DENOMINATOR RULE

Two denominators are used deliberately, both stated on the face of the artifact, so a reader reconciling the two figures does not conclude they contradict.

Breach rate — primary	33 of 160 = 20.6%. Past-due items over total inventory. The headline impact figure: how much of the current risk book is sitting in breach.
Breach rate — secondary	33 of 116 = 28.4%. Past-due items over the timeliness-eligible population — items whose due date has already passed. Answers a different question: of the items that have actually faced a deadline, what share failed to meet it. Carried as a footnote to the primary figure.
Eligibility note	44 items have due dates after the snapshot and cannot yet be late; 10 open, 34 already closed early. In the secondary view they count as neither successes nor failures — their outcome is still open. This prevents the primary figure from implicitly recording 44 pending items as satisfactory.
Severity metric	Overdue-to-SLA ratio, as defined in Q2. Reported with absolute days past due. Tier-independent by design.
Descriptive metrics	Volume, mix, control-effectiveness and domain distribution run on all 160. Only timeliness narrows to 116.

4 EVIDENCE DEVELOPED DURING FRAMING

CRITICAL	Cohort-over-time closure comparison is invalid. Closure rate by identification quarter runs 68, 72, 70, 73, 75, 77, 78 and 71 percent across eight quarters — essentially flat. Items identified as little as zero days before the snapshot show the same closure rate as two-year-old items, which is not possible in a real register. Status was assigned independent of item age. An eighth generator artifact; belongs in the Step 1 profile alongside tier, exposure and Days_Overdue.
CRITICAL	Apparent on-time improvement is right-censoring, not progress. On-time rate by cohort climbs from 85% to 100% while median cycle time falls from 119 days to 24. In recent cohorts only fast-closing items have had time to appear; slower ones remain open and absent from the calculation. The collapsing cycle time is the tell — that curve is an artifact of snapshot timing, not performance.
MATERIAL	SLA windows are severity-differentiated but overrun inverts. Median SLA: Critical 115 days, High 181, Medium 228, Low 263 — sensible. Median overrun against that window: Critical 2.94×, High 2.20× (max 4.90×), Medium 0.70×, Low 0.27×. The most severe items get the tightest deadlines and overrun them by the widest margin; 22 of 33 past-due items are now overdue longer than their entire original window. <i>Not yet verified</i> — must be tested in Step 4 against generator design before it can headline.
MATERIAL	The population bifurcates. Closed: 85% on time, worst case 59 days late, median cycle 106 days. Past due: mean 288 days overdue, median 305, max 523. Items either close roughly on time or become stuck for well over a year, with very little in between — the unresolved population is not a queue running late, it is a separate population behaving differently.
WATCH	Concentration exists but is too thin to name units. On the eligible denominator, breach rate ranges 17–43% across departments and 9–47% across owner roles, on base populations of only 13–25 items. Signal plausible, sample insufficient to attribute. Reportable as a pattern warranting further testing, not as an owner- or department-level conclusion.
WATCH	Breach rate is elevated at the severe end, but not monotonic. On the eligible denominator: Low 20.0%, Medium 23.3%, High 36.6%, Critical 33.3%. High and Critical sit clearly above Low and Medium, but Critical does not exceed High and the Low-to-Medium gap is negligible. The cleaner “rises with severity” reading produced by the all-160 denominator does not survive the eligibility correction — a worked example of why the denominator rule in Section 3 is enforced. Subject to the same Step 4 verification as the overrun inversion.

5 SCOPE BOUNDARY

In scope	What the data shows across all statuses; probable causes and areas warranting closer examination; explicit statement of uncertainty and of what the data cannot support.
Out of scope	Setting tolerance thresholds; designing controls or guardrails; prescribing remediation sequence or owner-level action. Senior leadership decisions, taken after findings are presented.
Not answerable	Trend, seasonality or change-point claims on volume or closure; cohort-over-time performance comparison; external or peer benchmarking; control concentration analysis; any exposure-to-tier relationship.

6 CARRY-FORWARD

To Step 3 — Analyze	Run the cross-sectional segment comparison on the defined denominators. Carry the Step 1 hypothesis forward: if residual score proves a mechanical haircut off inherent score, Control_Effectiveness does not drive risk reduction in this dataset.
To Step 4 — Verify	Adversarial priority is the tier-to-overdue relationship. If it is generator design rather than observed behaviour, severity inversion cannot headline and the fallback becomes the bifurcation itself, which does not depend on tier.
Housekeeping	Step 1 profile PDF requires regeneration to incorporate the flat-closure-rate finding. The 160-row dataset remains an unlogged open item in the Airtable Experiments table.

Source: grc_risk_register.csv (synthetic; no production or employer data). **Method:** Python / pandas — cohort closure and censoring tests, SLA-window and overrun-ratio computation, denominator eligibility reconciliation, segment concentration profiling. All figures computed, none estimated. **Confidence:** 95% on computed figures; 90% on the audit-committee action space as standard governance practice, noting individual committee charters vary; 75% on distinguishing generator artifacts from modelled business rules, which requires confirmation from the dataset author. **AI role (transparency):** analysis scripts and draft written by Claude (Opus 5) under human direction; framing decisions, scope boundary and findings reviewed and owned by N. Fragoso. **Prepared:** 30 Jul 2026.